

Assignment 10: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1
library(tidyverse); library(rvest); library(lubridate); library(viridis);
library(here); library(dataRetrieval); library(tidycensus)

here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):
 - Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
 - Scroll down and select the LWSP link next to Durham Municipality.
 - Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
theURL <-
  read_html("https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024")
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
- Water system name
- PWSID
- Ownership
- From the “3. Water Supply Sources” section:
- Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```
#3

water_system_name <- theURL %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()

PWSID <- theURL %>% html_nodes("td tr:nth-child(1) td:nth-child(5)") %>%
  html_text()

Ownership <- theURL %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()

MGD <- theURL %>% html_nodes("th~ td+ td") %>% html_text
```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It’s likely you won’t be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: “Jan”, “May”, “Sept”, “Feb”, etc. . .

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```

#4
water_supply_sources <- data.frame(
  "Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
  "Year" = rep(2024,12),
  "Water System Name" = water_system_name,
  "PWSID" = PWSID,
  "Ownership" = Ownership,
  "Max_Day_Use" = as.numeric(MGD)
)

#5

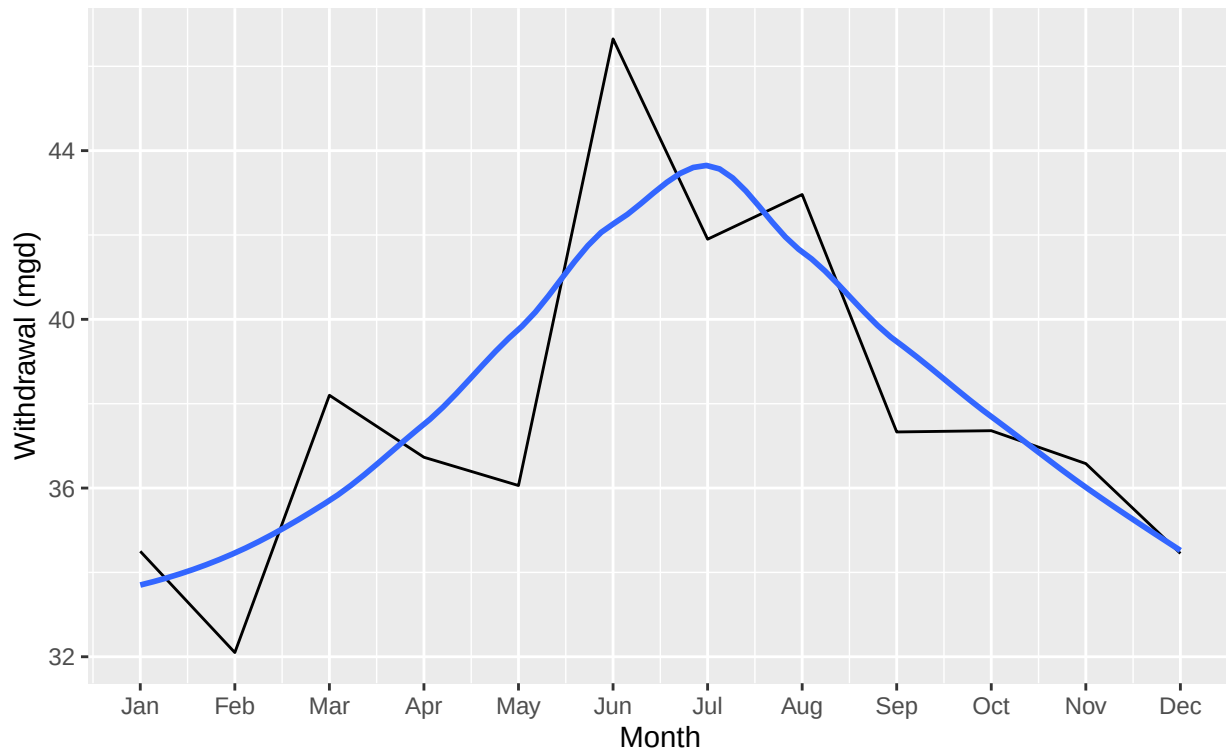
plot_Durham_2024 <- ggplot(water_supply_sources,aes(x=Month, y=Max_Day_Use)) +
  geom_line() +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  geom_smooth(method="loess",se=FALSE) +
  labs(title = paste(water_supply_sources$Year,
                     "Water Supply System", water_system_name),
       subtitle = paste("PWSID:", PWSID, "| Ownership:", Ownership),
       y="Withdrawal (mgd)",
       x="Month")
print(plot_Durham_2024)

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

2024 Water Supply System Durham

PWSID: 03-32-010 | Ownership: Municipality



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - “PWSID” and “year” - that:

- Creates a URL pointing to the LWSP for that PWSID for the given year
- Creates a website object and scrapes the data from that object (just as you did above)
- Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
- Returns the dataframe as the function’s output

```
#6.
the_base_url <- 'https://www.ncwater.org/WUDC/app/LWSP/report.php?psid='
the_PWSID <- '03-32-010'
the_year <- 2020
the_scrape_url <- paste0(the_base_url, '/', the_PWSID, '/', the_year)
print(the_scrape_url)

## [1] "https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=/03-32-010/2020"

scrape.it <-function(the_PWSID, the_year) {
  the_URL <-
    read_html(paste0('https://www.ncwater.org/WUDC/app/LWSP/report.php?psid=',
                     the_PWSID, '&year=', the_year))

  water_system_name <- the_URL %>%
    html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
    html_text()
  PWSID <- the_URL %>% html_nodes("td tr:nth-child(1) td:nth-child(5)") %>%
    html_text()
  Ownership <- the_URL %>%
    html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
    html_text
  MGD <- the_URL %>% html_nodes("th~ td+ td") %>% html_text

  water_supply_sources <- data.frame(
    "Month" = c(1,5,9,2,6,10,3,7,11,4,8,12),
    "Year" = rep(the_year, 12),
    "Water System Name" = water_system_name,
    "PWSID" = the_PWSID,
    "Ownership" = Ownership,
    "Max_Day_Use" = as.numeric(MGD)
  )
  return(water_supply_sources)
}
```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```
#7
plot_Durham_2020 <- scrape.it("03-32-010", 2020)

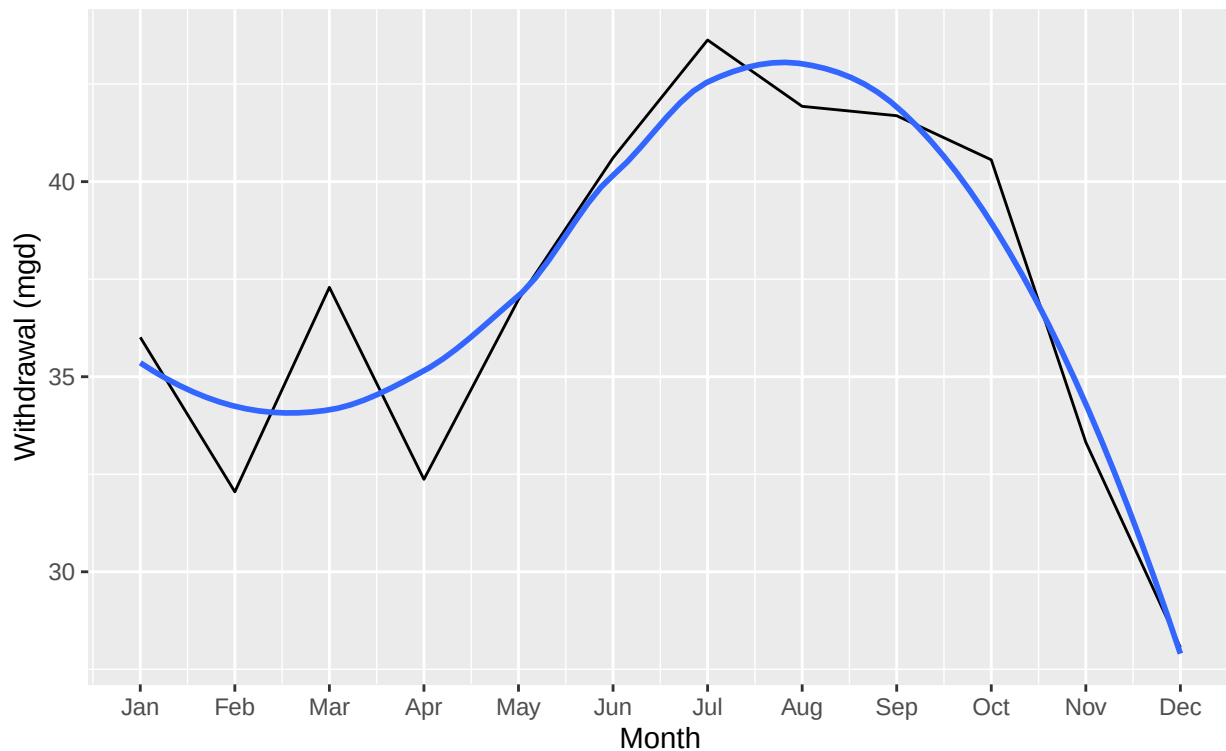
ggplot(plot_Durham_2020, aes(x=Month, y=Max_Day_Use)) +
```

```
geom_line() +
scale_x_continuous(breaks = 1:12, labels = month.abb) +
geom_smooth(method="loess",se=FALSE) +
labs(title = paste(plot_Durham_2020$Year, "Water Supply System in",
                    water_system_name),
      subtitle = paste("PWSID:", PWSID, "| Ownership:", Ownership),
      y="Withdrawal (mgd)",
      x="Month")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

2020 Water Supply System in Durham

PWSID: 03-32-010 | Ownership: Municipality



```
print(plot_Durham_2020)
```

##	Month	Year	Water.System.Name	PWSID	Ownership	Max_Day_Use
## 1	1	2020	Durham	03-32-010	Municipality	36.01
## 2	5	2020	Durham	03-32-010	Municipality	36.98
## 3	9	2020	Durham	03-32-010	Municipality	41.69
## 4	2	2020	Durham	03-32-010	Municipality	32.05
## 5	6	2020	Durham	03-32-010	Municipality	40.61
## 6	10	2020	Durham	03-32-010	Municipality	40.56
## 7	3	2020	Durham	03-32-010	Municipality	37.29
## 8	7	2020	Durham	03-32-010	Municipality	43.63
## 9	11	2020	Durham	03-32-010	Municipality	33.32
## 10	4	2020	Durham	03-32-010	Municipality	32.37

```
## 11      8 2020          Durham 03-32-010 Municipality      41.93
## 12     12 2020          Durham 03-32-010 Municipality      28.06
```

```
str(plot_Durham_2020)
```

```
## 'data.frame':   12 obs. of  6 variables:
## $ Month          : num  1 5 9 2 6 10 3 7 11 4 ...
## $ Year           : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ Water.System.Name: chr  "Durham" "Durham" "Durham" "Durham" ...
## $ PWSID          : chr  "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
## $ Ownership       : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Max_Day_Use     : num  36 37 41.7 32 40.6 ...
```

```
glimpse(plot_Durham_2020)
```

```
## Rows: 12
## Columns: 6
## $ Month          <dbl> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12
## $ Year           <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020
## $ Water.System.Name <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham", "Durham"
## $ PWSID          <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-32-010", "03-32-010"
## $ Ownership       <chr> "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality", "Municipality"
## $ Max_Day_Use     <dbl> 36.01, 36.98, 41.69, 32.05, 40.61, 40.56, 37.29, 43.~
```

8. Use the function above to extract data for Cary (PWSID = '03-32-010') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

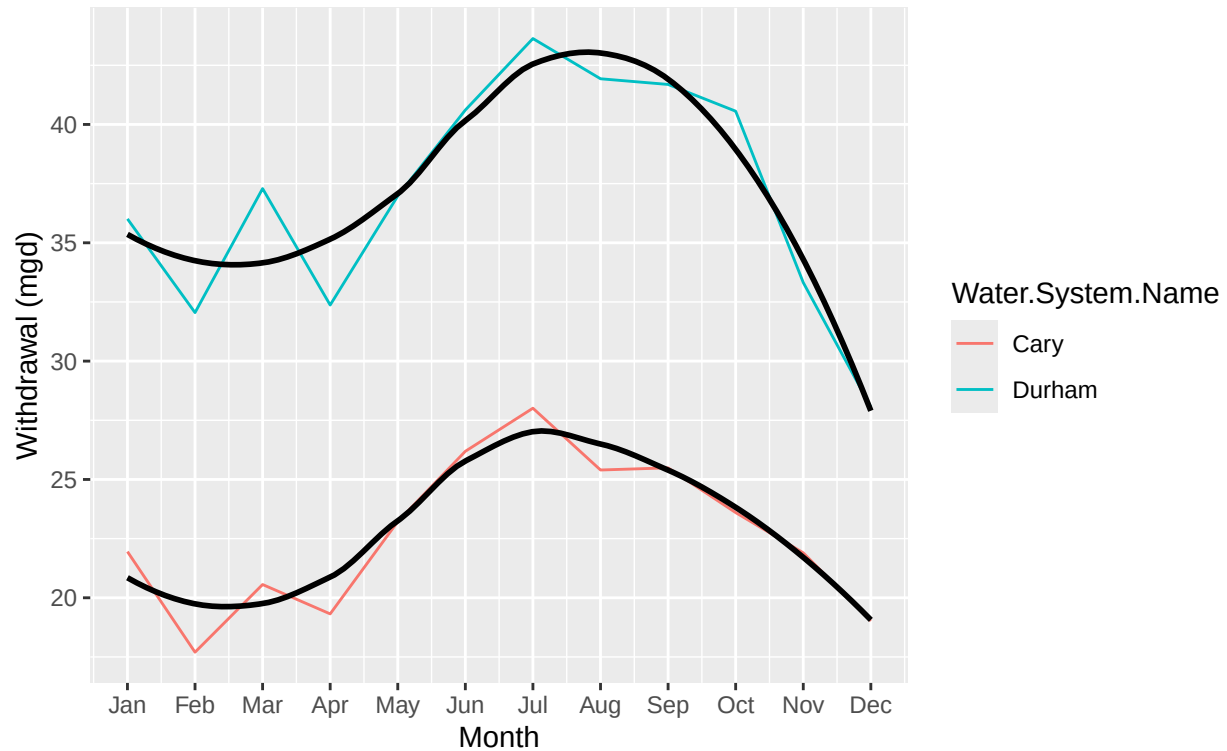
```
#8
plot_Cary_2020 <- scrape.it('03-92-020', 2020)

combined_2020 <- bind_rows(plot_Durham_2020, plot_Cary_2020)

ggplot(combined_2020, aes(x=Month, y=Max_Day_Use, group = Water.System.Name,
                          color = Water.System.Name)) +
  geom_line() +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  geom_smooth(method="loess", se=FALSE, color = "black") +
  labs(title = 'Comparison of Max Daily Withdrawals in 2020',
       subtitle = 'In Durham and Cary Counties',
       y="Withdrawal (mgd)",
       x="Month")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Comparison of Max Daily Withdrawals in 2020 In Durham and Cary Counties



```
print(combined_2020)
```

##	Month	Year	Water.System.Name	PWSID	Ownership	Max_Day_Use
## 1	1	2020	Durham	03-32-010	Municipality	36.01
## 2	5	2020	Durham	03-32-010	Municipality	36.98
## 3	9	2020	Durham	03-32-010	Municipality	41.69
## 4	2	2020	Durham	03-32-010	Municipality	32.05
## 5	6	2020	Durham	03-32-010	Municipality	40.61
## 6	10	2020	Durham	03-32-010	Municipality	40.56
## 7	3	2020	Durham	03-32-010	Municipality	37.29
## 8	7	2020	Durham	03-32-010	Municipality	43.63
## 9	11	2020	Durham	03-32-010	Municipality	33.32
## 10	4	2020	Durham	03-32-010	Municipality	32.37
## 11	8	2020	Durham	03-32-010	Municipality	41.93
## 12	12	2020	Durham	03-32-010	Municipality	28.06
## 13	1	2020	Cary	03-92-020	Municipality	21.95
## 14	5	2020	Cary	03-92-020	Municipality	23.22
## 15	9	2020	Cary	03-92-020	Municipality	25.49
## 16	2	2020	Cary	03-92-020	Municipality	17.70
## 17	6	2020	Cary	03-92-020	Municipality	26.19
## 18	10	2020	Cary	03-92-020	Municipality	23.60
## 19	3	2020	Cary	03-92-020	Municipality	20.56
## 20	7	2020	Cary	03-92-020	Municipality	28.01
## 21	11	2020	Cary	03-92-020	Municipality	21.90
## 22	4	2020	Cary	03-92-020	Municipality	19.32

```
## 23      8 2020          Cary 03-92-020 Municipality      25.40
## 24     12 2020          Cary 03-92-020 Municipality      19.02
```

```
str(combined_2020)
```

```
## 'data.frame':   24 obs. of  6 variables:
## $ Month          : num  1 5 9 2 6 10 3 7 11 4 ...
## $ Year            : num  2020 2020 2020 2020 2020 2020 2020 2020 2020 2020 ...
## $ Water.System.Name: chr  "Durham" "Durham" "Durham" "Durham" ...
## $ PWSID           : chr  "03-32-010" "03-32-010" "03-32-010" "03-32-010" ...
## $ Ownership        : chr  "Municipality" "Municipality" "Municipality" "Municipality" ...
## $ Max_Day_Use      : num  36 37 41.7 32 40.6 ...
```

```
glimpse(combined_2020)
```

```
## Rows: 24
## Columns: 6
## $ Month          <dbl> 1, 5, 9, 2, 6, 10, 3, 7, 11, 4, 8, 12, 1, 5, 9, 2, 6~
## $ Year            <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020, 2020~
## $ Water.System.Name <chr> "Durham", "Durham", "Durham", "Durham", "Durham", "D~
## $ PWSID           <chr> "03-32-010", "03-32-010", "03-32-010", "03-32-010", ~
## $ Ownership        <chr> "Municipality", "Municipality", "Municipality", "Mun~
## $ Max_Day_Use      <dbl> 36.01, 36.98, 41.69, 32.05, 40.61, 40.56, 37.29, 43.~
```

9. Use the code & function you created above to plot Cary’s max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = ‘loess’).

TIP: See Section 3.2 in the “10_Data_Scraping.Rmd” where we apply “map2()” to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

```
#9
the_years <- c(2018:2023)
Cary_PWSID <- rep.int('03-92-020', length(the_years))

dfs_Cary <- map2(Cary_PWSID, the_years, scrape.it)

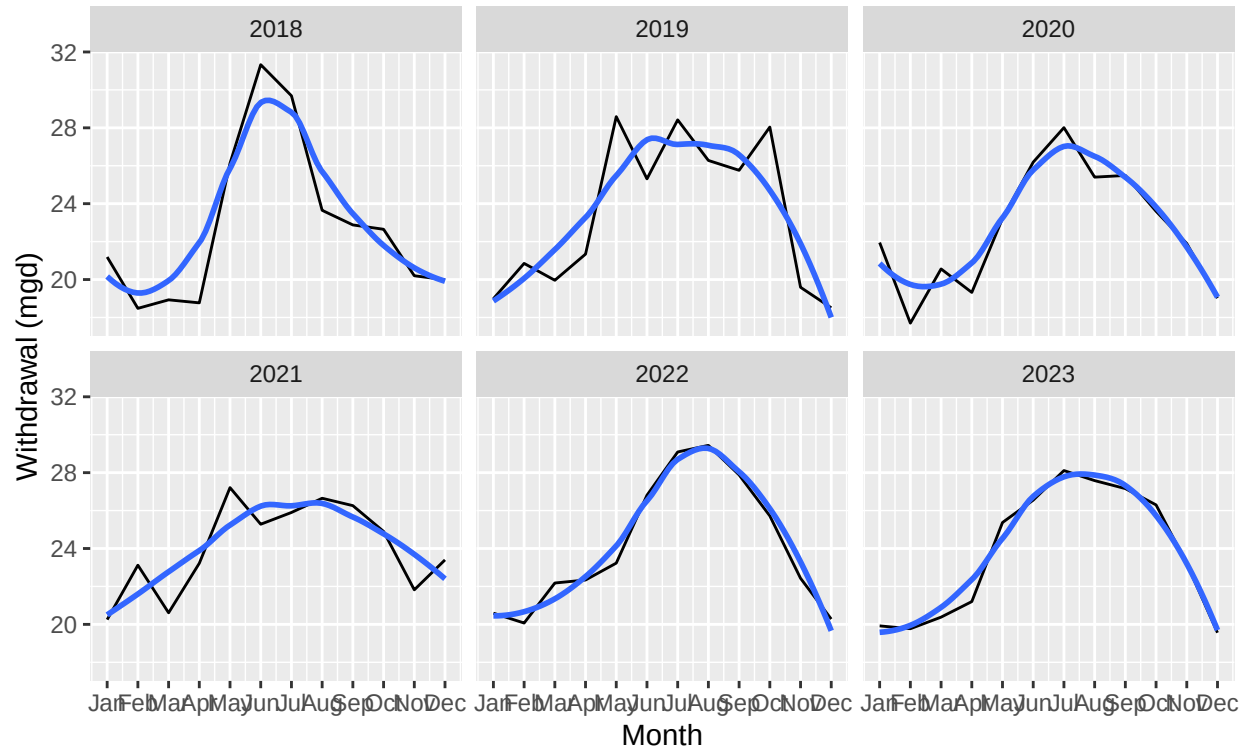
Cary_df <- bind_rows(dfs_Cary)

ggplot(Cary_df, aes(x=Month, y=Max_Day_Use)) +
  geom_line() +
  geom_smooth(method="loess", se=FALSE) +
  facet_wrap(~Year) +
  scale_x_continuous(breaks = 1:12, labels = month.abb) +
  labs(title = "Max Daily Withdrawals from 2018 to 2023 in Cary",
       subtitle = paste("PWSID:", Cary_df$PWSID, "| Ownership:",
                        Cary_df$Ownership),
       y="Withdrawal (mgd)",
       x="Month")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```


Max Daily Withdrawals from 2018 to 2023 in Cary

PWSID: 03-92-020 | Ownership: Municipality



```
print(Cary_df)
```

##	Month	Year	Water.System.Name	PWSID	Ownership	Max_Day_Use
## 1	1	2018	Cary	03-92-020	Municipality	21.19
## 2	5	2018	Cary	03-92-020	Municipality	26.13
## 3	9	2018	Cary	03-92-020	Municipality	22.88
## 4	2	2018	Cary	03-92-020	Municipality	18.48
## 5	6	2018	Cary	03-92-020	Municipality	31.33
## 6	10	2018	Cary	03-92-020	Municipality	22.65
## 7	3	2018	Cary	03-92-020	Municipality	18.93
## 8	7	2018	Cary	03-92-020	Municipality	29.69
## 9	11	2018	Cary	03-92-020	Municipality	20.20
## 10	4	2018	Cary	03-92-020	Municipality	18.77
## 11	8	2018	Cary	03-92-020	Municipality	23.65
## 12	12	2018	Cary	03-92-020	Municipality	19.99
## 13	1	2019	Cary	03-92-020	Municipality	18.99
## 14	5	2019	Cary	03-92-020	Municipality	28.59
## 15	9	2019	Cary	03-92-020	Municipality	25.76
## 16	2	2019	Cary	03-92-020	Municipality	20.85
## 17	6	2019	Cary	03-92-020	Municipality	25.31
## 18	10	2019	Cary	03-92-020	Municipality	28.04
## 19	3	2019	Cary	03-92-020	Municipality	19.96
## 20	7	2019	Cary	03-92-020	Municipality	28.42
## 21	11	2019	Cary	03-92-020	Municipality	19.59
## 22	4	2019	Cary	03-92-020	Municipality	21.34

## 23	8	2019	Cary	03-92-020	Municipality	26.28
## 24	12	2019	Cary	03-92-020	Municipality	18.51
## 25	1	2020	Cary	03-92-020	Municipality	21.95
## 26	5	2020	Cary	03-92-020	Municipality	23.22
## 27	9	2020	Cary	03-92-020	Municipality	25.49
## 28	2	2020	Cary	03-92-020	Municipality	17.70
## 29	6	2020	Cary	03-92-020	Municipality	26.19
## 30	10	2020	Cary	03-92-020	Municipality	23.60
## 31	3	2020	Cary	03-92-020	Municipality	20.56
## 32	7	2020	Cary	03-92-020	Municipality	28.01
## 33	11	2020	Cary	03-92-020	Municipality	21.90
## 34	4	2020	Cary	03-92-020	Municipality	19.32
## 35	8	2020	Cary	03-92-020	Municipality	25.40
## 36	12	2020	Cary	03-92-020	Municipality	19.02
## 37	1	2021	Cary	03-92-020	Municipality	20.25
## 38	5	2021	Cary	03-92-020	Municipality	27.21
## 39	9	2021	Cary	03-92-020	Municipality	26.26
## 40	2	2021	Cary	03-92-020	Municipality	23.12
## 41	6	2021	Cary	03-92-020	Municipality	25.28
## 42	10	2021	Cary	03-92-020	Municipality	24.90
## 43	3	2021	Cary	03-92-020	Municipality	20.61
## 44	7	2021	Cary	03-92-020	Municipality	25.90
## 45	11	2021	Cary	03-92-020	Municipality	21.82
## 46	4	2021	Cary	03-92-020	Municipality	23.22
## 47	8	2021	Cary	03-92-020	Municipality	26.65
## 48	12	2021	Cary	03-92-020	Municipality	23.40
## 49	1	2022	Cary	03-92-020	Municipality	20.59
## 50	5	2022	Cary	03-92-020	Municipality	23.23
## 51	9	2022	Cary	03-92-020	Municipality	27.88
## 52	2	2022	Cary	03-92-020	Municipality	20.07
## 53	6	2022	Cary	03-92-020	Municipality	26.81
## 54	10	2022	Cary	03-92-020	Municipality	25.72
## 55	3	2022	Cary	03-92-020	Municipality	22.18
## 56	7	2022	Cary	03-92-020	Municipality	29.09
## 57	11	2022	Cary	03-92-020	Municipality	22.43
## 58	4	2022	Cary	03-92-020	Municipality	22.34
## 59	8	2022	Cary	03-92-020	Municipality	29.44
## 60	12	2022	Cary	03-92-020	Municipality	20.28
## 61	1	2023	Cary	03-92-020	Municipality	19.92
## 62	5	2023	Cary	03-92-020	Municipality	25.37
## 63	9	2023	Cary	03-92-020	Municipality	27.16
## 64	2	2023	Cary	03-92-020	Municipality	19.77
## 65	6	2023	Cary	03-92-020	Municipality	26.55
## 66	10	2023	Cary	03-92-020	Municipality	26.30
## 67	3	2023	Cary	03-92-020	Municipality	20.38
## 68	7	2023	Cary	03-92-020	Municipality	28.11
## 69	11	2023	Cary	03-92-020	Municipality	23.20
## 70	4	2023	Cary	03-92-020	Municipality	21.20
## 71	8	2023	Cary	03-92-020	Municipality	27.58
## 72	12	2023	Cary	03-92-020	Municipality	19.56

Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Yes Answer: > When only looking at the annual trend of peak max daily withdrawals per month, we see peak withdrawals during the middle of each plot (during the warmer, summer months ranging from

June to August). When looking at the trend over the years, we see that the peak withdrawal used to begin earlier in the summer months (2018 peak withdrawal in June). This peak withdrawal has transition to later summer months (2023 peak withdrawal in around August).